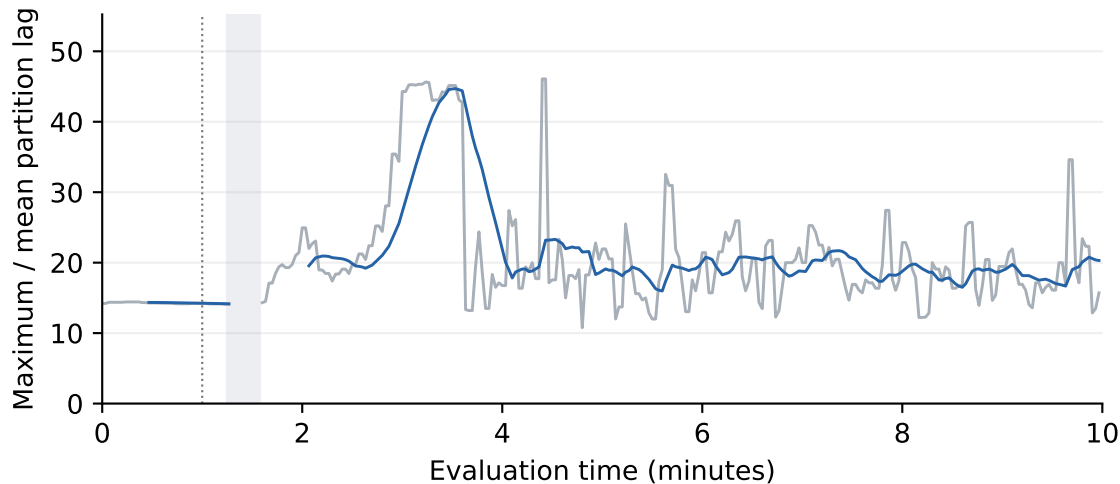


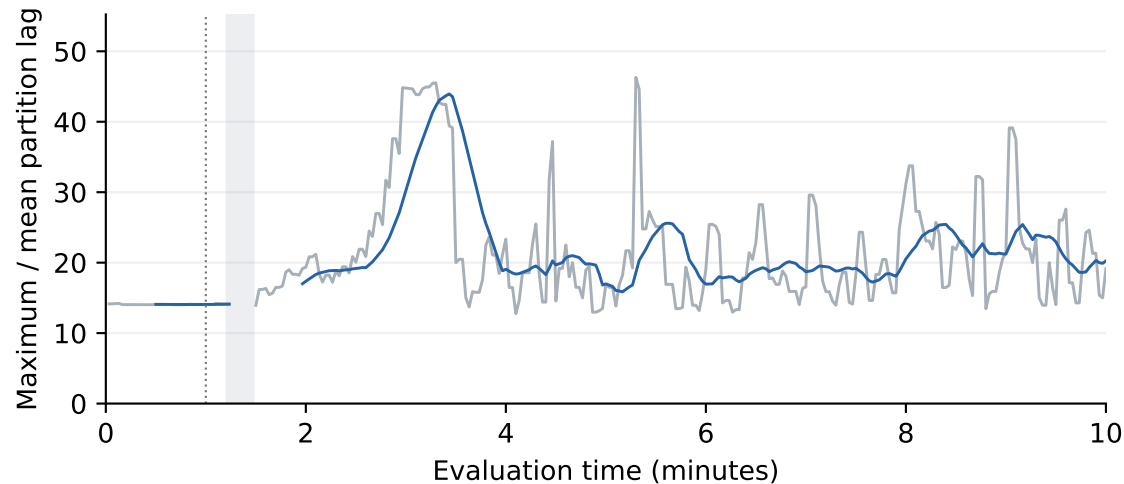
Partition lag skew - 700 messages/s

— Snapshot ratio — 15-sample mean

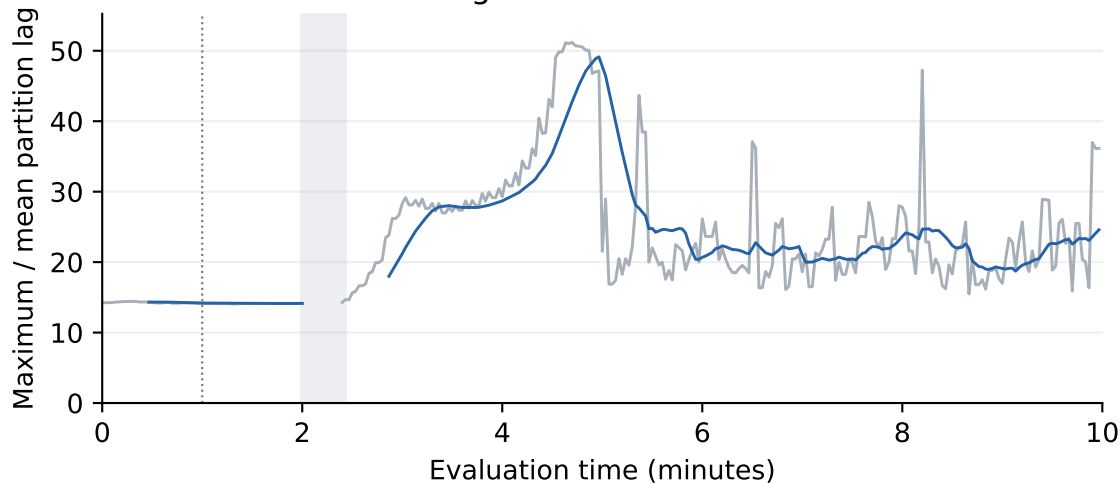
Redistribute within 3 - Run 1



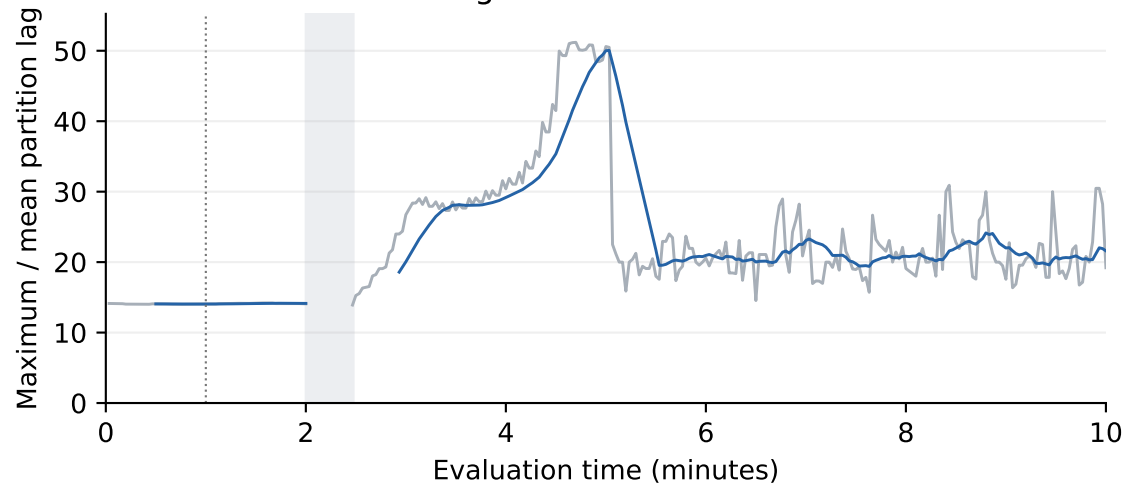
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2



80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.